REG # : L1F24BSCS0937 F25_COAL_C09_Assignment#1
~~L1F23SCS0000~~
NAME : M. Hussain Durrani.

This assignment will be marked on the basis of VIVA/Quiz-1.

INSTRUCTIONS:

1. Edit name and reg # in header before printing this assignment.
2. Then solve this assignment and scan using CamScanner.
3. Also add your neat working on white printer pages.
4. Submit L1F23BSCS0000.PDF File on portal.

Question#1

The values of various registers before executing each of the following instructions are as follows:

DS:0xA1B9 SI:0x0024 DI:0x0032 BP: 0x0110 SS:0x3C80

AX:0xABCD BX:0x12C2 CX:0xFEED DX:0x2032 ES:0x1AB0

Execute the following instructions and update the registers and memory accordingly. Also, calculate physical addresses in the hexadecimal number system. (10 Marks)

Instructions	Physical Address	Registers
MOV [BX+SI], CX	A2E76 h	
MOV [BP+DI+0x0016], DX	3C958 h	
MOV AX, [BX+DI]	A2E84 h	AX= 23 12
MOV [BX+SI+9D], CX	0x A2E7F	
MOV DX, DS:[BP+SI+11BCH]	0x A2E80	DX= F0 FE

Memory

			0x0	0x1	0x2	0x3	0x4	0x5	0x6	0x7	0x8	0x9	0xA	0xB	0xC	0xD	0xE	0xF
0x1AB0	:	0x0120																
0x1AB0	:	0x0130				50	56	78	AB									
0x1AB0	:	0x0140																
→ 0x3C80	:	0x0150								20	~~89~~ 32	20					AB	CD
0x3C80	:	0x0160																
→ 0xA1B9	:	0x12E0						ED	ED	~~10~~ FE	67						40	~~92~~ ED
→ 0xA1B9	:	0x12F0	FE ~~66~~	F0	50	30	12	23	45									
0xA1B9	:	0x1300															EE	89

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Question #2: Which of the following instructions is invalid? In the case of invalid, give reasons; otherwise, write "valid." (10 Marks)

Sr#	Instructions	Valid/Invalid	Reasons Based on Addressing Mode/ Logical Reason
1	MOV AX, BH	Invalid	Can't move 8 bit to a 16 bit Register.
2	MOV [SI+DI+4],256	Invalid	SI & DI can't come together. EA is invalid.
3	MOV [SI+BX+4], AX	Valid	
4	MOV ES, 0x1FCD	Invalid	Can't move immediate value directly into a segment register.
5	ADD DX, [BL+DI+2]	Invalid	BL not allowed. Only 'BX' allowed as base register.

Question#3 (3 + 3 + 3 Marks)

Write set of instructions to store 0x0BAD in memory at physical address 0x07DE1 using the following addressing modes one by one.

a) Direct Addressing Mode
b) Register Indirect Addressing Mode
c) Register relative base plus index addressing mode.

Part-A	Part-B	Part-C

MOV AX, 07D0h
MOV DS, AX
MOV WORD PTR [00E1H], 0BADh
	MOV AX, 07D0h
	MOV DS, AX
	MOV SI, 00E1h
	MOV WORD PTR [SI], 0BADh
		MOV AX, 07D0h
		MOV DS, AX
		MOV BX, 00E0h
		MOV SI, 00E1h
		MOV WORD PTR [BX+SI], 0BADh

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Question#4

Review the following instructions and fill in the table accordingly.

1. Write TRUE for VALID instructions only?
2. In the case of INVALID, write FALSE and also give reasons.

NOTE: Reasons based on rules or addressing modes. Avoid lengthy discussion.

Sr#	Instructions	Valid/ Invalid	Reasons for INVALID instructions
1.	MOV 0X1234, AX	False	Destination operand '0x1234' is an immediate constant. To make is memory or address, we should write: '[0x1234]'
2.	MOV AX, [CX]	False	CX is not a valid register for memory EA. Only BX, BP, SI, DI are allowed.
3.	MOV AL, [BL + 4]	False	BL (8-bit) can't be used in EA; only BX (16 bit) may be used.
4.	ADD DX, [BX+DX+2]	False	DX can't appear in offset. Only BX/BP as base & SI/DI as index are allowed.
5.	MOV CS, AX	False	The destination operand can't be a CS or IP register.
6.	ADD NUM1, NUM2	False	x86 doesn't allow memory-to-memory add. Atleast one operand must be a register.
7.	MOV DS, 0X720	False	Immediate value to a segment register is not Allowed.
8.	ADD AX, BL	False	Size Mismatch: 16 bit, 8 bit. Both operands should be the same size.
9.	MOV BL, SI	False	Size Mismatch: 8 bit, 16 bit. Should be the same size.
10.	MOV AX, [BP+BX]	False	'[BP+BX]' is not an allowed offset combination. It could be ⇒ [BP+SI] or [BP+DI] or [BX+SI/DI]
11.	MOV DX, [SI+DI]	False	SI+DI by itself is not allowed offset form. Possible forms → [ ]
12.	ADD BX, [NUMBER + CX]	False	CX is not a valid index register in offsets. Only BX/BP base & SI/DI index are allowed.
⇒ 13.	MOV [0X1234], BL	True	
⇒ 14.	MOV AX, [BX+SI+4]	True	

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F25_COAL_C-09_Assignment#1

Name: M. Hussain Durrani
Roll No: L1F24BSCS0937.

Q.NO.1

DS: 0xA1B9, SI: 0x0024, DI: 0x0032, BP: 0x0110, SS: 0x3C80
AX: 0xABCD, BX: 0x12C2, CX: 0xFEED, DX: 0x2032, ES: 0x1AB0

i) MOV [BX+SI], CX

Physical Address of '[BX+SI]' = Segment Register × 10 + Offset
= DS × 10 + [BX+SI]
= 0xA1B9 × 10 + 12C2h + 0024h
= A1B90h + 12C2h + 0024h = 12E6h → Offset
= A1B90h + 12E6h

Physical Address = A2E76h

⇒ Logical Address = Segment Address : Offset
= DS: [BX+SI]
Logical Address = A1B9: 12E6 → Put value of CX at this address.

Memory ⇒

ED	FE
0x5	0x6

ii) MOV [BP+DI+0x0016], DX

Physical Address of '[BP+DI+0016h]' = SS×10 + [BP+DI+0016h]
= 3C80h × 10 + 0110h + 0032h + 0016h
= 3C800 + 0110 + 0032 + 0016 = 0158h → Offset
= 3C800 + 0158

Physical Address = 3C958h

⇒ **Logical Address = 3C80: 0158** → Put value of DX at this address.

Memory ⇒

32	20
0x	0x

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iii) MOV AX, [BX+DI]

Physical Address of '[BX+DI]' = DS×10 + [BX+DI] = A1B9h×10 + 12C2h + 0032

= A1B90h + $\boxed{12F4h}$ → Offset

$$= \begin{array}{r} A1\overset{1}{B}90h \\ +12F4h \\ \hline \end{array}$$

$\boxed{\text{Physical Address} = A2E84}$

⟹ $\boxed{\text{Logical Address} = A1B9:12F4}$ → Put the value here in AX.

Memory ⟹

12	23
0x4	0x5

AX = | 23 | 12 |

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iv) MOV [BX+SI+9D], CX

Physical Address of '[BX+SI+9D]' = DS×10 + [BX+SI+9D]

= A1B9h×10 + 12C2h + 0024h + 9D

$$= A1B90h + \begin{array}{r} 12C2 \\ +0024 \\ \hline 12E6 \end{array} + 9D$$

$$= A1B90h + \begin{array}{r} 12E6 \\ +0009 \\ \hline \boxed{12EF} \end{array}$$ → Offset

$$= \begin{array}{r} A1\overset{1}{B}90h \\ +12EFh \\ \hline \end{array}$$

$\boxed{\text{Physical } \cancel{\text{Logical}} \text{ Address} = A2E7Fh}$

⟹ $\boxed{\text{Logical Address} = A1B9:12EF}$ → Put value of CX at this Address.

Memory ⟹

ED	FE
12EFh	12F0h

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v) MOV DX, DS:[BP+SI+11BCh]

Physical Address of 'DS:[BP+SI+11BCh]' = DS×10 + [BP+SI+11BCh]

= A1B9h×10 + 0110h + 0024h + 11BCh

$$= A1B90h + \begin{array}{r} 01\overset{1}{3}4h \\ +11BCh \\ \hline \boxed{12F0} \end{array}$$ → Offset

$$= \begin{array}{r} A1\overset{1}{B}90h \\ +12F0h \\ \hline \end{array}$$

$\boxed{\text{Physical Address} = A2E80}$

⟹ $\boxed{\text{Logical Address} = A1B9:12F0}$ → Put the value here in DX

Memory ⟹

FE	F0
12F0h	12F1h

DX = | F0 | FE |

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